ANALYSIS AND SYSTEMS OF BIG DATA | COURSE PROJECT

Player Replacement Scouting

Mining top-club replacements from lower-table talent across Europe's top five football leagues

Mid-Semester Report | October 2026

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At a glance

Big Data · Knowledge discovery summary

Question. Can data mining find *young players at non-elite clubs* who play like a current starter of an elite club, and do those players later move up? We answer it with seven seasons (2017-18 to 2023-24) of player data from the five big European leagues, joined with team-style data and league tables, and we judge every method against the simple base rate of "pick a random young player".

18,243 raw player rows from 5 leagues x 7 seasons, read in one DuckDB query	**16** per-90 metrics per player, weighted for team style and league	**13 + 10** syllabus techniques up to classification, plus 10 beyond the syllabus	**2.8x** best hit rate over a random scout in unseen seasons

Most important findings so far (real data; details in section 5):

- A data-driven shortlist works: players who look like a top-club starter moved up **18.3%** of the time against **8.3%** for all young players (permutation p = 0.006), and the best methods found **11 movers in 45 picks** against about 4 for a random scout.
- But nothing yet **clearly beats simply picking the best young performers** (25.0%); the gap is inside the noise. Similarity adds explainability more than accuracy.
- Team style really changes a player's numbers (49 of 96 effects proven); weighting for it **cut style bias by 55%** and makes players from different systems comparable.
- **Playing time** (starter share, AUC 0.70) is the strongest single clue that a club will buy a player; FP-Growth and Apriori return identical patterns, and the mined rules help only modestly out of time (1.22x).
- The outcome definition was the biggest weakness: **40%** of young players were labelled "left the top 5 leagues", and moves to top clubs outside the Premier League were never counted. Refining it is the first task for the second half of the semester.
- The project covers **every syllabus topic up to classification**, including all seven frequent-itemset algorithms written from scratch and a full decision-tree study, and adds **10 techniques beyond the syllabus** (section 3).
- Early reality check: of the 13 players named on the 2023-24 list, **at least four** have since joined elite clubs (Leny Yoro, Bryan Mbeumo, João Pedro, Illia Zabarnyi).

SCOPE OF THIS REPORT

This mid-semester report covers the pipeline and the analyses completed so far; every number comes from the full run on the real Kaggle data. Section 6 lists the current limitations and section 7 the work planned for the second half of the semester, starting with extending the elite reference and the definition of "moving up" to all five leagues.

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1 Problem and objective

Big Data · Data mining for decision support

Football clubs spend most of their transfer budget replacing first-team players. A club at the top of a league wants a younger, cheaper player who can do the same *job* as an existing starter; a scout can only watch a handful of matches a week. The project turns this into a data-mining question: given the statistical profile of every player in the five big leagues, which young players at non-elite clubs look like elite starters, which combinations of traits precede a move to a bigger club, and how reliable is each method when it is tested on seasons it has never seen?

Objectives

- Build a reproducible big-data pipeline that joins player, team-style and league-table data across 5 leagues and 7 seasons.
- Make player numbers comparable across tactical systems and leagues before any mining (team-style and league weights).
- Apply the full range of course techniques: descriptive statistics, pre-processing, feature extraction, dimensionality reduction, clustering, outlier analysis, frequent-pattern and association-rule mining, similarity measures, classification and regression.
- Validate honestly: base rates, out-of-time tests, walk-forward evaluation, confidence intervals and a simple "best performers" control.
- Produce a short, explainable scouting list for the latest season.

Definitions used throughout. A **feeder** is a player aged 24 or under with at least 900 minutes at a non-elite club. An **elite starter** has at least 1,350 minutes at an elite club. A player **moves up** if, two seasons later, he plays for an elite club or makes a step up in league. The **hit rate** is the share of picked players with a positive outcome, and **lift** is a hit rate divided by the base rate of the whole feeder pool.

2 Data and pipeline

Big Data · Columnar SQL (DuckDB) · data integration

Source (Kaggle)	Content	Use in the project
akshankrithick / fbref-2017-2024-for-europes-top-5-leagues	FBref player season tables, 2017-18 to 2023-24, 5 leagues	player profiles (16 per-90 metrics), age, minutes, club
mexwell / understat-database	Understat match-level team data: xG, deep completions, PPDA, points	team style (dominance, pressing, strength); league tables for all 5 leagues
evangower / english-premier-league-standings	official final Premier League tables	defines the elite reference clubs (Premier League top 3 of each season)

Everything is fetched with the Kaggle CLI and read by **DuckDB** in one query over all season files (*union_by_name*), so new seasons only need a new file. Club names differ between sources, so an alias table plus an accent-insensitive "skeleton" key matched **145 clubs one-to-one**; the run fails if any club is unmatched.

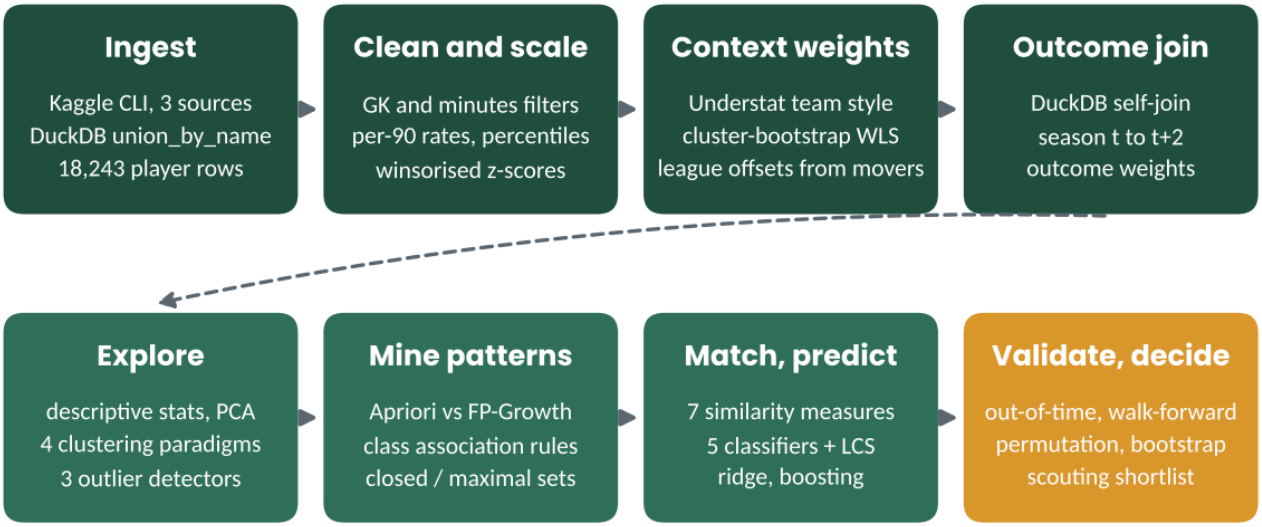


Figure 1. The pipeline. Every stage is a set operation over the whole multi-season table rather than a per-club loop, which is what lets it scale.

Pre-processing audit

Data pre-processing

Step	Rows left	Removed	Reason
Raw rows loaded (5 leagues x 7 seasons)	18,243	0	union of all season files
Goalkeepers dropped	16,865	1,378	outfield per-90 metrics do not describe goalkeepers
At least 10 full matches (900 minutes)	10,435	6,430	per-90 rates from tiny samples are noise
One main row per player-season	10,355	80	a mid-season transfer creates two rows; keep the one with most minutes
Labelled (season t+2 exists in the data)	7,385	2,970	the outcome can only be scored if two seasons later is observed
Eligible feeder player-seasons	1,452	5,933	age, minutes and club filters

No raw metric cells were missing; blanks would mean "no recorded action" and are filled with 0. Four requested metrics (shots on target, xAG, touches, recoveries) are absent from this dataset. Skewed metrics were handled three ways: log transform (median skewness 0.97 to 0.26), winsorising at the 1st/99th percentile within position, and percentile ranks within season and position (skewness 0.00), which also remove position and season effects.

3 Techniques used

Big Data · Course syllabus coverage

The project is required to cover the course topics up to classification and to go beyond what was taught in class. The last column of the table marks each technique as **taught** (part of the syllabus covered so far) or **beyond** (implemented in addition to it). Every technique is in the notebook because it changes a later decision, not as decoration.

Technique	Methods	Purpose	Headline result	Course
Descriptive statistics	mean, median, IQR, skewness, kurtosis, box plots, correlation	scale and shape of 16 metrics by position	median skewness 0.97; gaps up to 11.7x	Taught
Data pre-processing	filters, de-duplication, winsorising, percentiles, z-scores, discretisation	clean and comparable inputs	audit table, nothing dropped silently	Taught
Feature extraction and selection	8 engineered football features, univariate ROC-AUC	football concepts; rank single signals	starter share best (AUC 0.70)	Taught
Dimensionality reduction	PCA (scree, loadings, projection)	compress 16 correlated metrics	7 components keep 80%	Taught
Frequent itemsets	Apriori vs FP-Growth; closed and maximal itemsets	patterns of traits; cost and compression	identical results; maximal sets 34%	Taught
Course itemset algorithms	hash-based Apriori (DHP), DIC, vertical Eclat, Partition, A-Close, Pincer-Search, written from scratch	compare scans, candidates and runtime on player data	all outputs identical to FP-Growth	Taught
Association rules and interestingness	support, confidence, lift, leverage, conviction, cosine, Jaccard, Kulczynski	trait combinations that precede a move up	13 trustworthy rules, best lift 7.6x	Taught
Similarity measures	cosine, Pearson, Euclidean, Manhattan, Mahalanobis on 3 representations	find look-alikes of elite starters	18.3% hit rate vs 8.3% pool	Taught
Inferential statistics	Wilson intervals, chi-square, Mann-Whitney, Fisher exact, Cohen's h	uncertainty around every claim	shortlist vs pool p = 0.006	Taught
Classification	decision tree, naive Bayes; ROC-AUC, PR-AUC, precision@K, confusion matrix	predict who moves up	best 11/45 picks (2.8x random)	Taught
Decision tree induction	Gini vs entropy vs log-loss, depth, leaf size, cost-complexity pruning, tree trace	how the tree is built and why a player is picked	336 trees x 3 unseen seasons	Taught
Learning classifier system	two-stage bucket brigade with credit assignment	rule-based learner	6/45, AUC 0.51	Taught
Prediction (regression)	ridge regression vs persistence and mean baselines	performance two seasons ahead	R-squared 0.35, beats persistence	Taught
Columnar SQL engine	DuckDB: union of all season files, self-joins, window functions	ingest; join season t to t+2 for every player	18,243 rows in one query	Beyond
Entity resolution	alias table + accent-insensitive skeleton key	match clubs across FBref and Understat	145 clubs matched, 0 left over	Beyond
Regression with resampling	minutes-weighted least squares, cluster bootstrap	how team style changes each position's output	49 of 96 effects supported	Beyond
Natural experiment	players who changed league act as their own control	league weights relative to the Premier League	467 movers, 39 of 64 offsets	Beyond
Clustering (4 paradigms)	K-Means, Ward hierarchical, Gaussian mixture, DBSCAN; silhouette, DB, CH, ARI	team tactical families; player archetypes	4 team styles, 9 archetypes	Beyond
Outlier analysis	robust Mahalanobis, Isolation Forest, Local Outlier Factor	unusual profiles; are they bought more?	no link to moving up (p = 0.78)	Beyond

Technique	Methods	Purpose	Headline result	Course
Multiple-testing control	Benjamini-Hochberg false discovery rate over all candidate rules	keep only trustworthy rules	18 candidates to 13 rules	**Beyond**
Resampling inference	permutation test, bootstrap confidence intervals	is the shortlist better than chance or a simple control?	p = 0.006; CI -21.7% to +8.3%	**Beyond**
Ensemble and linear models	random forest, gradient boosting, logistic regression	stronger classifiers and regressors	logistic regression 11/45	**Beyond**
Validation design	out-of-time split, walk-forward evaluation, ablation study	no future information in any test	all results on unseen seasons	**Beyond**

Table. 13 techniques from the syllabus up to classification and 10 techniques beyond it.

4 Findings

Big Data · Inference from the mined results

4.1 Team style and league context

Regression + cluster bootstrap

A raw per-90 number mixes the player with his team. Understat data describes every club-season by territorial dominance (share of deep completions), pressing (PPDA) and strength (non-penalty xG difference). Two independent sources agree on how much of the game a team controls (Spearman 0.82 between an FBref passing proxy and Understat dominance). K-Means on these variables finds four tactical families in 684 club-seasons: Low-block / reactive (217), High-press (176), Balanced / direct (168) and Possession-dominant (123); the Premier League has the largest low-block share (41%).

Largest supported style effects	Elasticity	Reading
Forwards: passes into the box vs dominance	+0.37	about 37% more per unit of extra dominance
Midfielders: passes into the box vs dominance	+0.30	dominant teams feed the box through midfield
Forwards: carries into the box vs dominance	+0.25	more possession, more box entries
Midfielders: progressive passes vs dominance	+0.23	progression is partly the system
Defenders: progressive passes vs pressing	+0.22	pressing sides push defenders up
Defenders: tackles won vs pressing	+0.21	pressing creates tackles

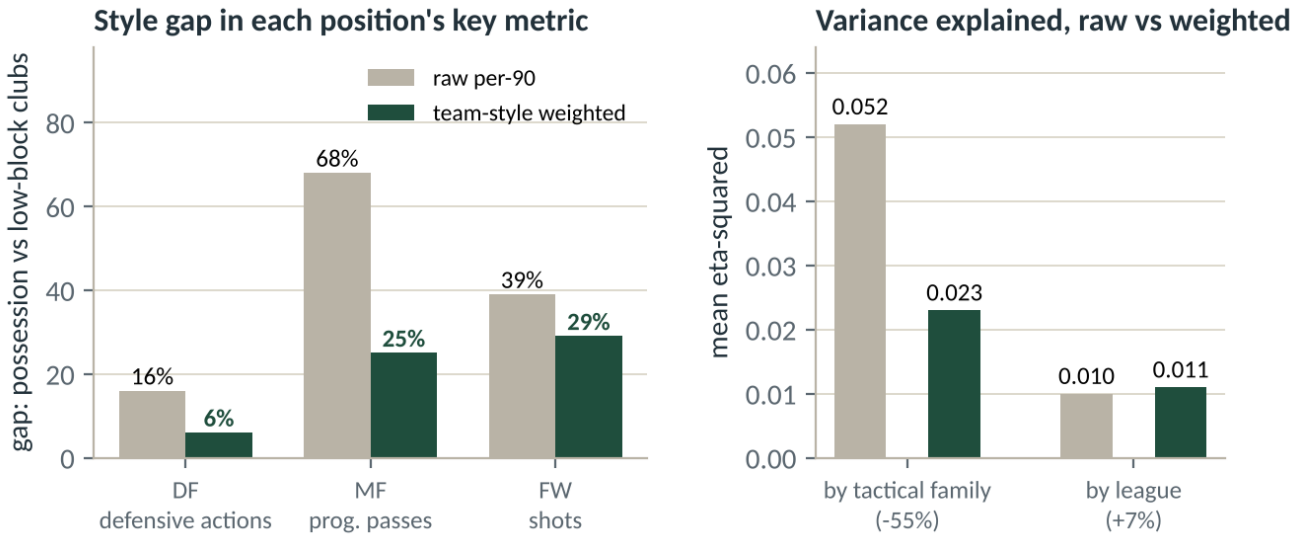


Figure 2. Left: gap between possession-dominant and low-block clubs in each position's key metric, before and after weighting. Right: variance explained by tactical family and by league (eta-squared).

KEY TAKEAWAY

Team style changes what players "do" in half of the tested cases and differently for each position. After weighting, the gap between possession and low-block clubs shrinks for all three positions (midfielders from 68% to 25%) and style bias falls by 55% (39 of 48 metrics improved). League weights from 467 players who changed league were statistically supported for 39 of 64 metrics but did not reduce league variance, which was already small (0.010).

4.2 Who moves up? Base rates and the outcome definition

SQL self-join · label definition

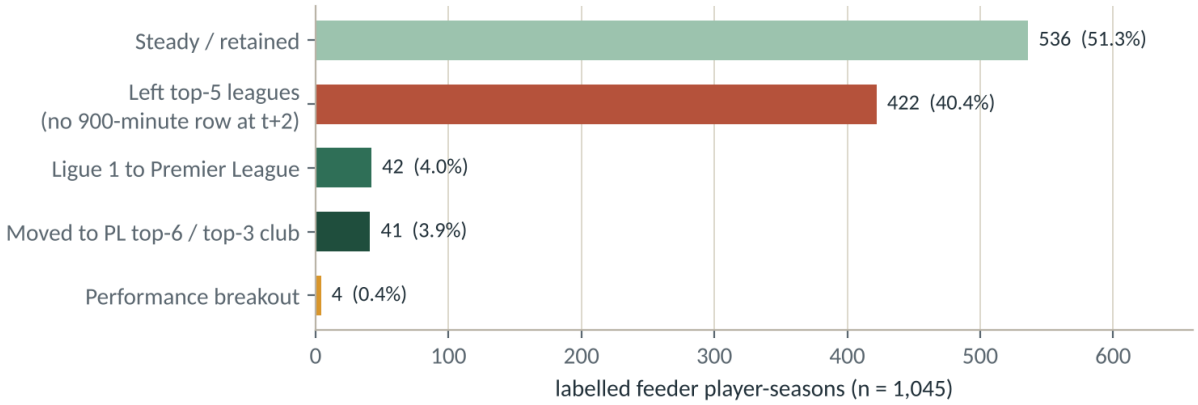


Figure 3. Outcome two seasons later for the 1,045 labelled feeders (current outcome rule).

Under the current rule a young feeder had an **8.3%** chance of a positive outcome (6.4% to 10.6% depending on the season) and a **3.9%** chance of joining a Premier League top-six or top-three club. These base rates are the bar every method must clear.

WEAKNESS FOUND

The red bar is the largest weakness found so far. The current rule looks only at the player's main club two seasons later and only if he played 900+ minutes there, so a young signing who sat on the bench at a big club, or who moved to Bayern, Real Madrid or Inter, is counted as "left the top 5 leagues" or "steady". The refined rule looks at every row in the next two seasons at any minutes, in all five leagues (section 6).

4.3 Descriptive statistics, feature extraction and PCA

Descriptive statistics · PCA

- **Shape:** most metrics have long right tails (median skewness 0.97, kurtosis up to 6.0 for carries into the box); goals are exactly zero for 28% of players.
- **Position matters:** the median of carries into the box differs 11.7x between defenders and forwards, so every trait is ranked within position and season.
- **Redundancy:** 20 of 120 metric pairs correlate above 0.7 (key passes vs shot-creating actions 0.94), so far fewer than 16 independent dimensions exist.
- **Feature selection:** the strongest single signals of a future move are starter share (AUC 0.70), aerial win % (0.69), progressive passes (0.65) and the composite index (0.63).

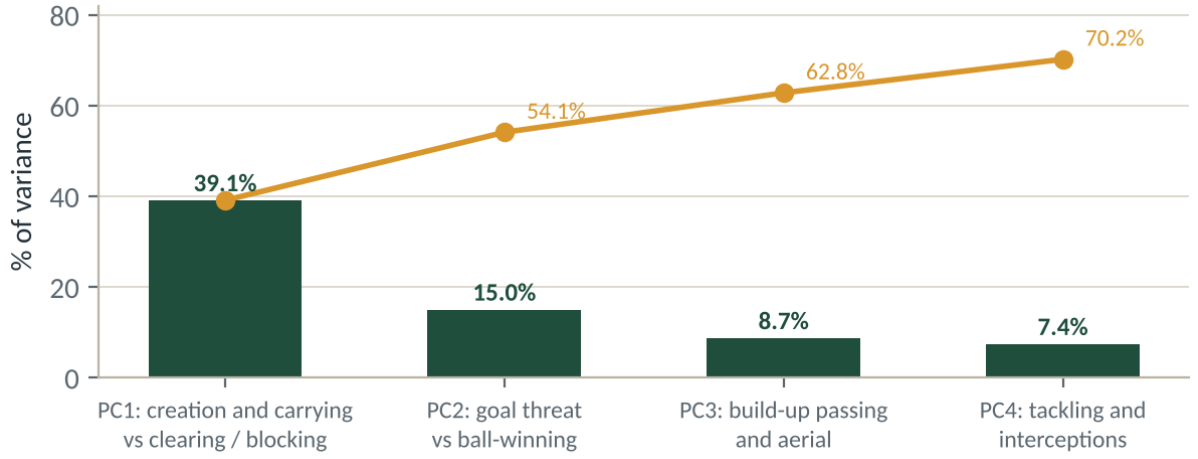


Figure 4. Variance explained by the first four principal components (bars) and cumulatively (line); 7 components keep 80% and 10 keep 90%.

KEY TAKEAWAY

PC1 is overall attacking involvement for the role. Players who later moved up sit between ordinary feeders and top-3 starters on this axis (PC1 0.21 vs 0.07 vs 1.21): clubs buy players who are already moving towards the elite profile, not finished copies of it.

4.4 Clustering and outlier analysis

Clustering · outlier detection

Inside each position, players were clustered in PCA space with four paradigms. K-Means gave the best internal quality (silhouette 0.21 to 0.28, k = 3 per position); Ward agreed well (ARI 0.48 to 0.73); the Gaussian mixture preferred 7 to 8 components but with lower silhouette (about 0.11); DBSCAN found one dense core plus 4% to 6% noise. The low silhouettes say players form a **continuum** of styles rather than separate types.

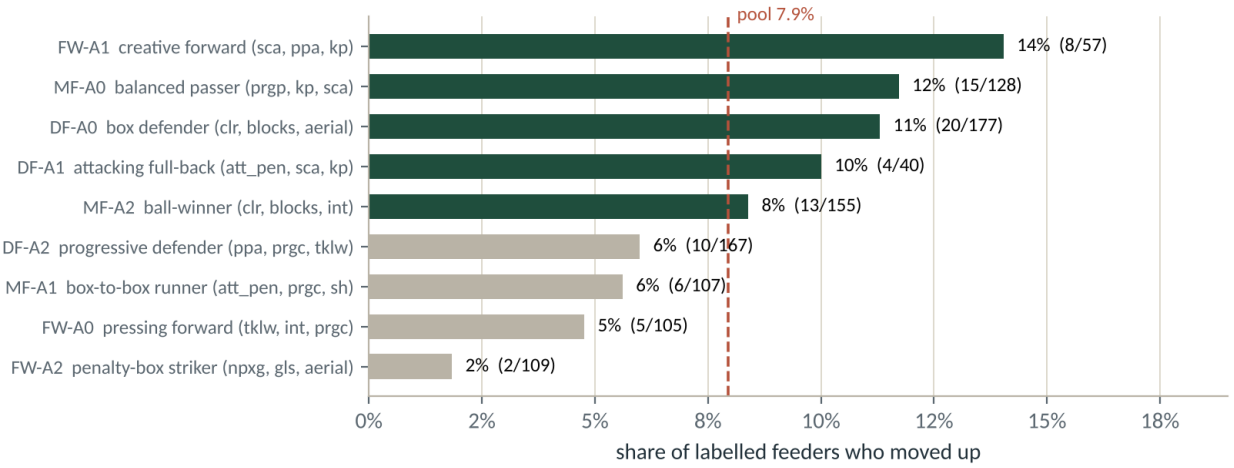


Figure 5. Share of feeders in each archetype who moved up (labels interpreted from each cluster's three strongest traits; dashed line = pool rate).

KEY TAKEAWAY

Archetype and moving up are linked (chi-square p = 0.029, Cramer's V 0.13): creative forwards (14%), balanced passing midfielders (12%) and box defenders (11%) are bought most; penalty-box strikers least (2%). Three outlier detectors agreed only partly (mean Jaccard 0.20) and an unusual profile is not more likely to be bought (Spearman -0.009, p = 0.78): outliers are leads for video scouting, not recommendations.

4.5 Frequent patterns and association rules

Apriori · FP-Growth · rule mining

Each feeder season is a transaction of 27 yes/no items: a "high" flag (top quartile for position and season) for each of 16 traits, a top-quartile composite flag, position, age band, league and two outcome items. Rules were mined on 581 seasons up to 2019 and tested on 464 later seasons.

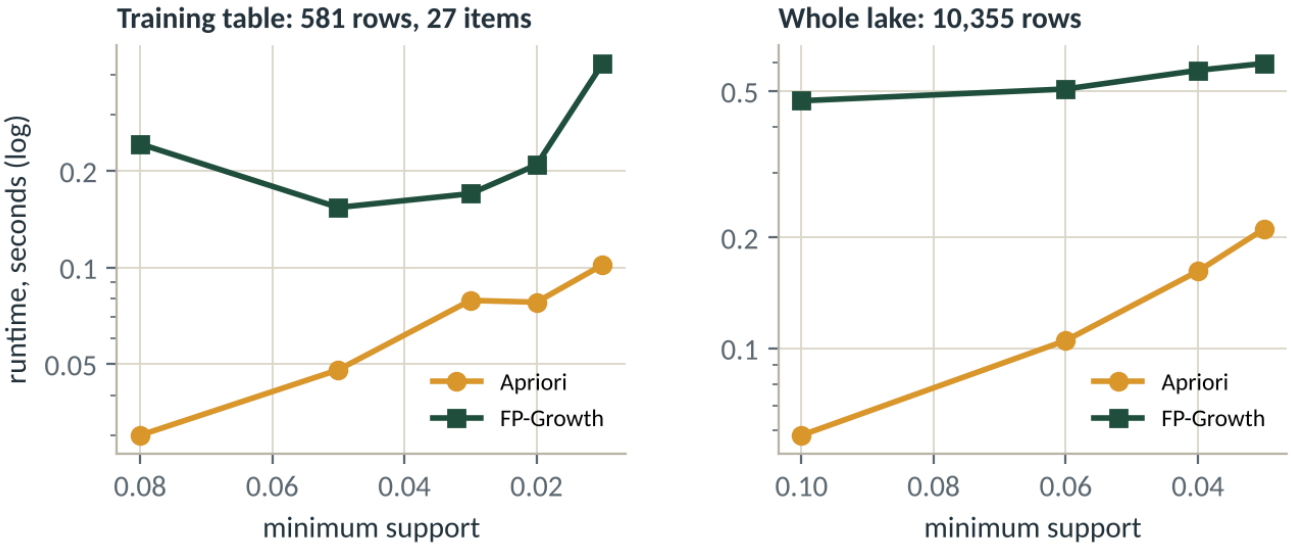


Figure 6. Runtime of Apriori and FP-Growth as the support floor drops. Both returned identical itemsets at every level.

On these modest table sizes Apriori was faster (4.3x on the training table, 2.8x on the 10,355-row lake) because FP-Growth pays a tree-building overhead; the gap narrows as the table grows, which is the scaling argument for FP-Growth on very large data. 6,564 frequent itemsets produced 18 candidate rules backed by at least 8 players, and 13 survived lift, Benjamini-Hochberg FDR and redundancy pruning.

Rule (antecedent)	Outcome	Players	Confidence	Lift
top-quartile composite + high aerial	top club	9	25.0%	7.6x
top-quartile composite + Premier League	top club	8	15.7%	4.8x
age 21-22 + high carries + Ligue 1	Ligue 1 to PL	8	15.4%	4.5x
top-quartile composite + high carries + Ligue 1	Ligue 1 to PL	10	14.1%	4.1x
age 21-22 + high progressive carries	Ligue 1 to PL	8	11.0%	3.2x
high progressive carries + Ligue 1	Ligue 1 to PL	12	9.2%	2.7x
high aerial win %	top club	13	8.7%	2.7x
top-quartile composite	top club	12	8.2%	2.5x

Min. support	Frequent	Closed (lossless)	Maximal (lossy)	Longest itemset
0.10	182	182 (100%)	96 (53%)	4
0.07	448	448 (100%)	207 (46%)	5
0.05	918	916 (99.8%)	336 (37%)	6
0.04	1,559	1,541 (98.8%)	525 (34%)	7

KEY TAKEAWAY

Being good overall and strong in the air is the clearest route to a top club (lift 7.6x), and progressive carrying is the Ligue 1 to Premier League signature. Closed itemsets compress almost nothing (99% of the set, verified lossless) because these traits rarely share identical support; maximal itemsets cut it to a third. Out of time, players matching a rule moved up 11.6% vs 9.5% for all test feeders (lift 1.22x, p = 0.074): positive but not statistically significant.

4.6 The course frequent-itemset algorithms

DHP · DIC · Eclat · Partition · A-Close · Pincer

Library Apriori and FP-Growth were complemented by implementing every frequent-itemset algorithm of the course from scratch and running them on the same player transactions (the 27 trait, context and outcome items, with the outcome items removed). Each algorithm answers the same question in a different way, so the comparison is about cost: how many full passes over the data, how many candidate itemsets have to be counted, and how long it takes.

Algorithm	How it works	Passes over the data	Output
Apriori (reference)	level-wise join and prune; one scan per itemset size	longest itemset + 1	all frequent itemsets
Hash-based Apriori (DHP)	pairs hashed into buckets in the first scan; a pair is a candidate only if its bucket is frequent; transactions trimmed	as Apriori, far fewer candidate pairs	all frequent itemsets
Dynamic Itemset Counting (DIC)	data read in blocks; new candidates start at any block boundary once all their subsets look frequent	fewer than one full pass per level	all frequent itemsets
Vertical mining (Eclat)	item to set of transaction ids (bit masks); support = size of an intersection; depth-first	one	all frequent itemsets
Partition	each chunk mined locally; the union of local results verified in one more scan	exactly two	all frequent itemsets
A-Close	level-wise search for generators, then closure = intersection of the transactions that contain them	levels + 1	frequent closed itemsets (lossless)
Pincer-Search	bottom-up levels plus a top-down maximum frequent candidate set counted in the same scans	often fewer than Apriori	maximal frequent itemsets

Verification. The notebook asserts that the frequent itemsets and their exact counts from Apriori, DHP, DIC, Eclat and Partition are identical to the library FP-Growth, that A-Close returns exactly the closed itemsets and Pincer-Search exactly the maximal itemsets found in section 4.5, at three support levels on the training table and once on the 10,355-row lake. On a test dataset with the same structure all seven agreed with the reference; Eclat needed one pass, Partition two, DIC 2.5 to 3.25, and the hash filter removed up to 90% of the candidate pairs before the second scan.

RESULTS

The full-data runtimes, pass counts and candidate counts for this comparison are produced by the notebook (section 7c3) and will be added to this report once the full run is complete.

4.7 Similarity shortlist and inferential statistics

Similarity search · inference

Each season, every top-3 starter's standardised 16-metric profile was compared by cosine similarity with every feeder of the same position; each starter contributes his closest feeder and the 12 best matches above 0.85 form the shortlist. A same-size list of the best performers by composite index is the control.

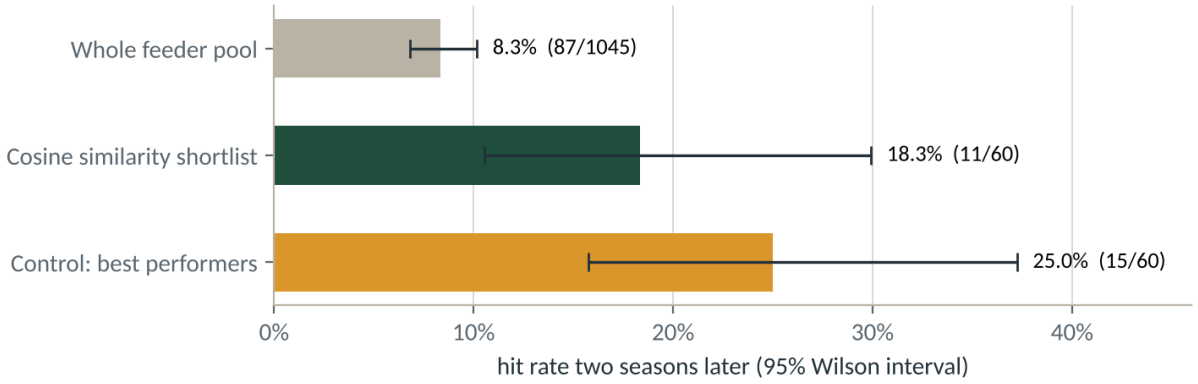


Figure 7. Hit rate two seasons later, pooled over five cohorts, with 95% Wilson intervals.

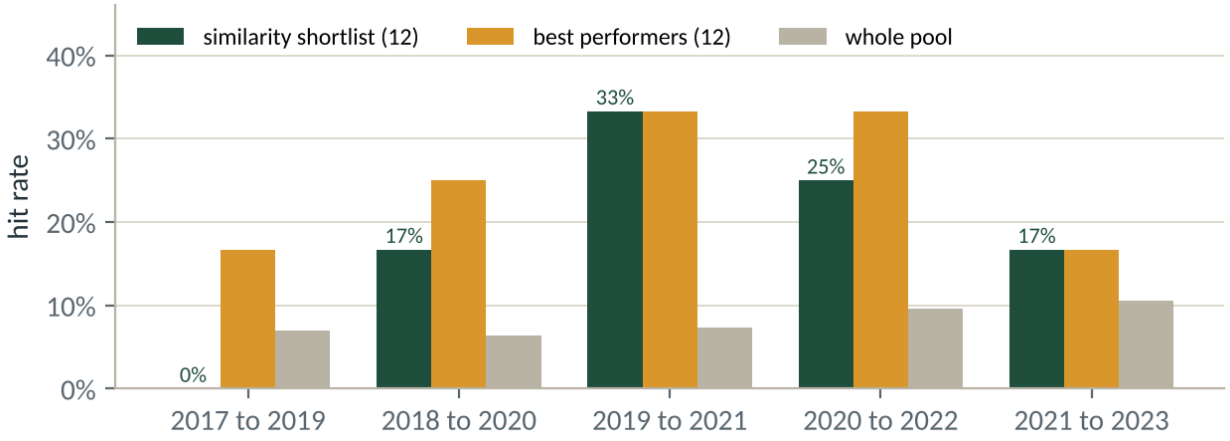


Figure 8. The same comparison cohort by cohort; with 12 players per list, single cohorts are noisy.

Player (season, club)	Matched elite starter	Similarity	Two seasons later
Diogo Jota (2018, Wolves)	Raheem Sterling (Man City)	0.943	Liverpool
Gabriel Magalhães (2018, Lille)	Virgil van Dijk (Liverpool)	0.957	Arsenal
William Saliba (2020, Nice)	Rúben Dias (Man City)	0.958	Arsenal
Jack Grealish (2019, Aston Villa)	Riyad Mahrez (Man City)	0.873	Manchester City
James Maddison (2021, Leicester)	Mason Mount (Chelsea)	0.955	Tottenham
Benoît Badiashile (2021, Monaco)	Joël Matip (Liverpool)	0.950	Chelsea
Sven Botman (2020, Lille)	Harry Maguire (Man Utd)	0.926	Newcastle (Ligue 1 to PL)

Table. Shortlisted players who did move up (examples from the real-data run).

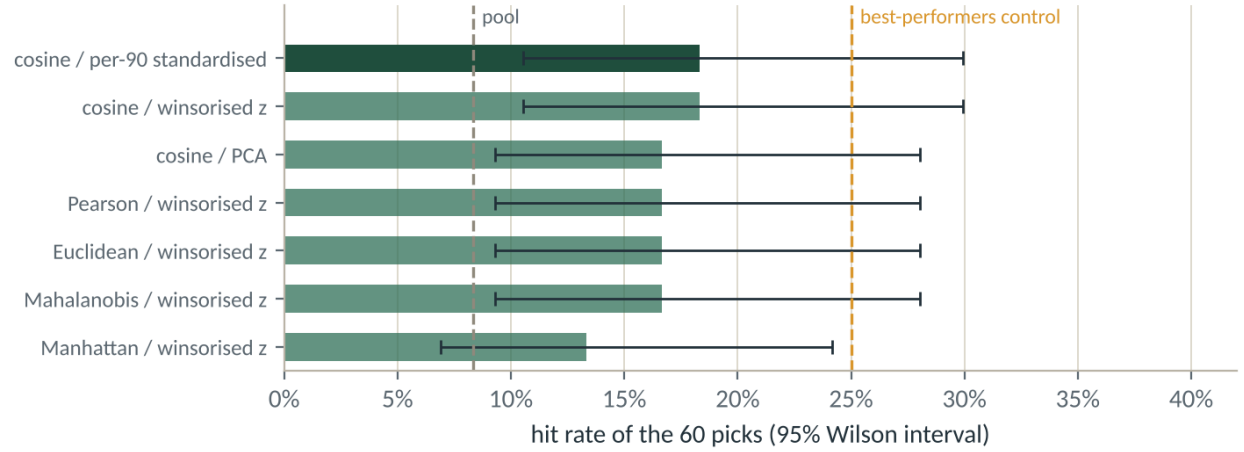


Figure 9. Seven similarity measures under identical rules (60 picks each).

KEY TAKEAWAY

The shortlist beats random scouting clearly (18.3% vs 8.3%; permutation p = 0.006, Cohen's h 0.30), but not the simple control: shortlist minus best performers is -6.7% with a 95% bootstrap interval of -21.7% to +8.3%. All seven similarity measures land between 13% and 18% and are statistically indistinguishable, yet they pick different players (only 31% overlap). Movers differ from stayers mainly in minutes (median 2,352 vs 1,826, AUC 0.70) and composite performance (54.7 vs 50.6).

4.8 Supervised prediction

Classification · walk-forward validation

Eight methods were trained only on seasons whose outcome was already known and asked to pick 15 players in each of three unseen seasons (2019 to 2021). In those seasons 8.7% of feeders moved up, so a random scout would expect about 4 hits in 45 picks.

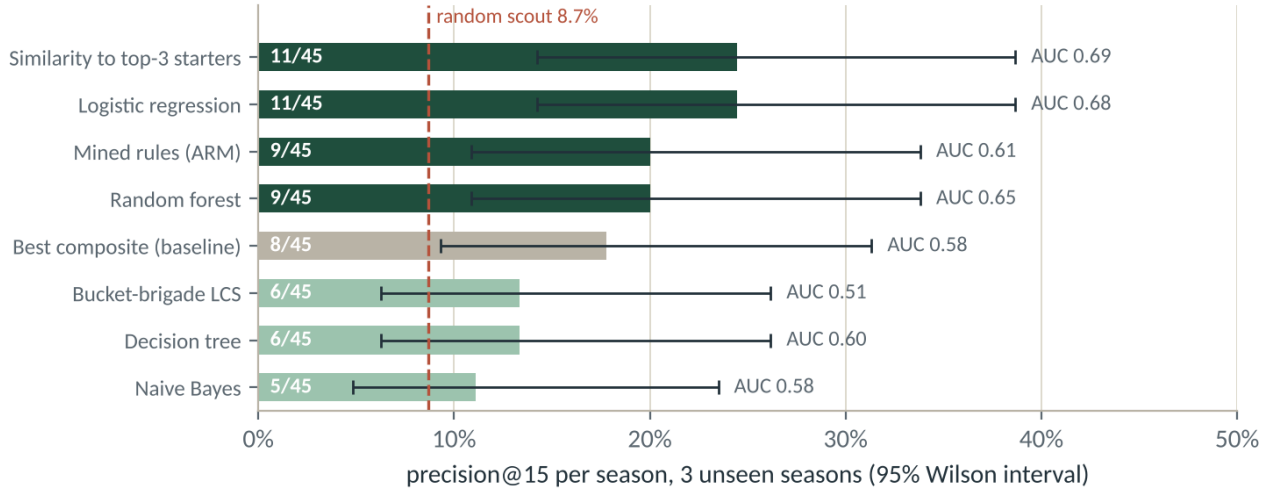


Figure 10. Hits among each method's 45 picks with 95% Wilson intervals; AUC = ranking quality over all thresholds.

Features	Model	Mean AUC	Hits / 45
raw per-90 percentiles	Logistic regression	0.716	13
raw per-90 percentiles	Random forest	0.738	13
team-style + league weighted	Logistic regression	0.700	14
team-style + league weighted	Random forest	0.721	11

Table. Ablation: the same models on raw versus weighted features.

Regression model (index two seasons ahead)	MAE	RMSE	R-squared
Ridge regression	6.94	8.63	0.35
Gradient boosting	7.01	8.71	0.34
Persistence ("same as now")	7.59	9.50	0.21
Mean of training targets	8.81	10.70	0.00

Table. Regression on 270 unseen player-seasons; ridge beats persistence by 0.65 points (95% CI 0.10 to 1.22).

KEY TAKEAWAY

Five of eight methods beat a random scout (binomial $p < 0.05$); similarity to top-3 starters and logistic regression did best (11/45, 2.8x). Their intervals overlap, so the order among methods is not reliable, and none clearly beats ranking by current performance (8/45). The weights make players fairer to compare but do not improve prediction (AUC about -0.017), because clubs also buy on raw output and reputation. Development is partly predictable (R-squared 0.35), with clear regression to the mean.

4.9 Decision tree study and the trace of the tree

Decision tree induction

The leaderboard used a single fixed tree (Gini, depth 3, at least 15 players per leaf), which reached a mean AUC of 0.60 and 6 hits in 45 picks. The tree study examines how the tree is built, with the same walk-forward protocol and the same 40 features:

- **Split criteria:** Gini impurity, entropy (information gain) and log-loss.
- **Hyperparameters:** maximum depth 2 to 8 and unlimited; minimum samples per leaf 5, 15, 30 and 60; cost-complexity pruning (ccp_alpha 0 to 0.01).
- **Scoring:** 336 configurations x 3 unseen seasons = 1,008 trees, each scored by mean ROC-AUC, average precision and hits among the top picks; the training AUC is kept to expose over-fitting.
- **Trace:** the best tree is printed as if-then rules and drawn, and the exact root-to-leaf path of three real test players (a mover it picked, a mover it missed, a non-mover it picked) is shown step by step with the share of similar training players who moved up at every node.

Trace of the current tree (real data). Trained on all labelled seasons, the fixed tree reads as three scouting rules:

Path	Condition	Prediction
1	aerial win % percentile above 70 and progression percentile above 22 (not a penalty-box striker archetype)	moved up
2	aerial win % percentile 70 or below, starter share above 0.48 and tackles-won percentile above 19	moved up
3	any other path (for example starter share 0.48 or below)	stayed

Table. The depth-3 tree from the real-data run, read from root to leaf.

KEY TAKEAWAY

Even the small tree turns into readable scouting conditions: strong in the air and progressive, or a regular starter who wins tackles. On the test dataset the study showed the expected pattern: the three criteria differ by less than 0.01 AUC (entropy and log-loss build identical trees in scikit-learn), while an unlimited tree climbs to a training AUC near 0.9 but falls on new seasons, so depth and pruning matter far more than the split criterion. The full-data numbers come from notebook section 9A-3.

4.10 The 2023-24 scouting list and an early reality check

Decision support

All models were re-fitted on every labelled season and applied to the 198 eligible feeders of 2023-24. A **priority target** needs both role fit (on the similarity shortlist) and a model probability in the top quartile.

Player	Club (age)	Matched elite starter	P(move up)	Status
Arthur Theate	Rennes (23)	Rúben Dias	0.22	priority
Jan Paul van Hecke	Brighton (23)	Ibrahima Konaté	0.18	priority
Illia Zabarnyi	Bournemouth (20)	Gabriel Magalhães	0.18	priority; **joined PSG, Aug 2025**
Kevin Danso	Lens (24)	Virgil van Dijk	0.17	priority
Andy Pelmard	Clermont Foot (23)	Manuel Akanji	0.14	priority
Toti Gomes	Wolves (24)	William Saliba	0.11	priority
Bryan Mbeumo	Brentford (23)	Mohamed Salah	0.11	priority; **joined Man Utd, Jul 2025**
Anthony Gordon	Newcastle (22)	Bukayo Saka	0.30	priority
João Pedro	Brighton (21)	model only	0.41	**joined Chelsea, Jul 2025**
Leny Yoro	Lille (17)	model only	0.32	**joined Man Utd, Jul 2024**

Table. Priority targets and two model-only names. Transfers are from public reports (sources listed in the appendix), not yet from the dataset.

KEY TAKEAWAY

Within two seasons at least four of the thirteen named players joined elite clubs, including Zabarnyi to PSG, a move the current rule cannot count because PSG is outside the Premier League. This is anecdotal (no base-rate comparison yet), but it is exactly the evidence the five-league outcome rule planned for the second half will measure properly.

5 Most important findings

Big Data · Knowledge discovery

1. **Scouting by data beats chance, clearly.** Look-alikes of elite starters moved up 2.2x as often as random young players, and the best learned models reached 2.8x on unseen seasons.
2. **Quality and minutes are most of the signal.** Starter share and the composite index are the strongest single clues; no method has yet proven better than "pick the best young performers".
3. **Context must be removed before comparing players.** Team style shifts per-90 numbers by position (49 of 96 effects); weighting cut style bias by 55%.
4. **Some trait combinations are real signals.** Overall quality plus aerial strength (lift 7.6x) and progressive carrying in Ligue 1 (4.1x) precede moves, but rules transfer to new seasons only weakly.
5. **Players form a continuum, not types.** Clusters are weak (silhouette at most 0.28), yet archetypes still differ in how often they are bought (chi-square p = 0.029).
6. **Algorithms agree; cost depends on scale.** Apriori and FP-Growth always return identical itemsets; on these sizes Apriori is faster, and the gap narrows as the table grows.
7. **Honest validation changes conclusions.** Without the best-performers control the similarity method would look like a breakthrough; with it, the gain is within noise.
8. **The label is the weakest link.** The current rule mislabels 40% of players and ignores moves to elite clubs outside the Premier League; fixing it is the first priority for the second half.

6 Current limitations and planned refinements

Big Data · Data quality audit

Area	Current approach	Planned refinement (second half)
Reference ("whom do we replace?")	starters of the Premier League top 3 only, about 30 profiles a season, mostly two clubs	starters of the top-N clubs of each of the five leagues (default N = 6), about 10x more profiles across many systems
Who can be scouted	under-25s at Premier League and Ligue 1 clubs	under-25s at every non-elite club of all five leagues
What counts as moving up	PL traditional six or PL top 3, or Ligue 1 to PL	any top-N club or traditional giant of any league, graded: top-4 or giant 1.0, places 5-6 0.75, stronger league 0.5, breakout 0.25
How the move is detected	main club at t+2 only, and only with 900+ minutes there	best destination in t+1 or t+2 at any minutes; age check to avoid name clashes (DuckDB window query)
League tables	official PL table only	tables for all five leagues rebuilt from Understat match rows, validated against the official PL table
Top 3 vs top 6	fixed at top 3	sensitivity analysis: top 3, 4 and 6, Premier League only vs all five leagues

Other known limitations:

- Outcomes are inferred from the club a player appears for in later seasons, not from a transfer-history feed with fees and dates.
- Aerial win % is a raw percentage, so players with few duels can look extreme; shrinkage towards the position mean is planned.
- Positions are coarse (DF, MF, FW); a centre-back and a full-back are compared as one group.
- Consecutive cohorts share players and only three test seasons exist, so intervals are wide and p-values somewhat optimistic.
- The list of traditional giants is fixed by hand, and league tables rebuilt from match data will not include point deductions.

7 Plan for the second half of the semester

Big Data · Project roadmap

Phase	Work	Deliverable
1. Five-league extension	Use the top clubs of all five leagues as the reference and as destinations, apply the refined outcome rule, and choose top 3, 4 or 6 from the sensitivity analysis.	corrected labels, updated findings
2. Better data	Add seasons 2024-25 and 2025-26; a transfer-history source for fees and dates; shrinkage for rate metrics; finer positions (CB, FB, DM, CM, AM, W, ST).	richer, more accurate labels
3. Scale and serve	Store the lake as partitioned Parquet; show the same queries in PySpark to demonstrate scaling; an interactive dashboard for the scouting list.	scalable pipeline, dashboard
4. Final evaluation	Re-run every analysis, including the course itemset algorithms and the decision tree study, on the corrected labels with the same walk-forward protocol; write the final report.	final report and presentation

FOCUS FOR THE SECOND HALF

The goal of the second half is not more methods for their own sake but a fairer test: correct labels across all five leagues, a reference set chosen by data, and every analysis re-run on those labels. Only then can we say whether similarity or learning really beats picking the best young performers.

8 Conclusion

Big Data · Summary

The project shows that a big-data pipeline over five leagues and seven seasons can turn raw player statistics into a short, explainable scouting list that finds future movers far more often than chance. It also shows where the value really lies: removing team and league context makes players comparable, honest validation keeps claims in proportion, and the definition of "moving up" matters as much as any algorithm. In the second half of the semester we will widen the reference and the destinations to all five leagues, repair the outcome label, and re-run every analysis on the corrected labels.

A Glossary and sources

Big Data · Reference

per-90	a season total divided by minutes / 90	**precision@K**	share of hits among the top K picks
percentile	rank within season and position, 0 to 100	**lift**	rate in a group divided by the base rate
composite index	mean percentile over the 16 metrics	**FDR**	false discovery rate (Benjamini-Hochberg)
npxG	non-penalty expected goals	**eta-squared**	share of variance explained by a grouping
PPDA	opponent passes allowed per defensive action (lower = more pressing)	**Wilson interval**	95% confidence interval for a proportion
deep completion	a pass completed within about 20 m of goal	**ARI**	adjusted Rand index; agreement between two clusterings
starter share	minutes divided by 3,420 (a full season)	**LCS**	learning classifier system (bucket brigade)
AUC	area under the ROC curve; 0.5 = coin flip		

Transfer sources for section 4.10

- Leny Yoro to Manchester United (July 2024): ewn.co.za/2024/07/19/manchester-united-sign-18-year-old-french-defender-yoro
- Bryan Mbeumo to Manchester United (July 2025): tntsports.co.uk, "Manchester United confirm deal to sign Bryan Mbeumo from Brentford"
- João Pedro to Chelsea (July 2025): thepeninsulaqatar.com/article/02/07/2025/chelsea-sign-brazil-striker-joao-pedro-from-brighton
- Illia Zabarnyi to Paris Saint-Germain (August 2025): aljazeera.com/sports/2025/8/12/illia-zabarnyi-psg-sign-ukraine-defender-from-bournemouth

Shadow Squad: who replaces whom?

End-semester report, Player Replacement Scouting
Analysis and Systems of Big Data, IIITDM Kancheepuram

This part follows the mid-semester report, which is included unchanged before it. Every number below is read from `outputs/metrics.json`, which the notebooks `endsem_01` to `endsem_06` write when they run.

1 What changed since the mid-semester version, and why

The mid-sem project asked which young players at smaller clubs will move up. Its main result was that no method clearly beat ranking the best young performers, and its cosine shortlist was worse than that baseline. Three things were limiting it, and the end-sem version changes all three.

The question. A club does not buy "a player who will move up". It buys a replacement for a specific starter who left. The end-sem system takes an elite club, a season and a departing starter, and returns a ranked list of candidates. The test is whether the player the club really signed is near the top of that list.

The labels. Mid-sem outcomes were inferred from the club a player's name appeared next to one or two seasons later. They are now taken from Transfermarkt transfer records. On the 1,953 feeder-seasons that have both labels, the two agree on "moved up" for 90.8%. The old rule called 149 moves that have no matching permanent transfer; most are loan spells and loan returns, and 23 belong to players with no transfer record in the Transfermarkt dump. On the same rows the move-up rate falls from 26.6% to 20.5%. The transfer label also covers 2022-23 and 2023-24, which the old rule could not label.

The data. Transfermarkt (`davidcariboo/player-scores`) adds transfers, fees, market values, fine positions and a stable player id. FBref removed its advanced statistics in January 2026, so the seasons stay 2017-18 to 2023-24. A Kaggle scrape with 2024-25 exists (`emrey3lmaz`) but was not merged; Transfermarkt already supplies the 2024-25 minutes and transfers that the outcomes need.

The mid-sem notebook was re-run unchanged first. The minutes column it uses is named `avg_mins_per_match` but holds season totals (maximum 3420), so no fix was needed. Its baseline on the refined labels: feeder base rate 26.7%; best walk-forward model, logistic regression, 46 hits in 75 picks; best-composite baseline 31 in 75.

2 Data lake and PySpark pipeline

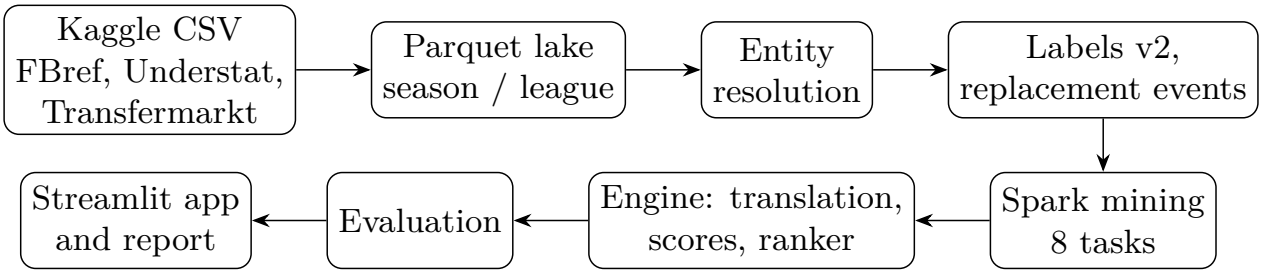


Figure 1: Pipeline. Every box after the first reads and writes Parquet under `lake/`.

All sources are written as Parquet partitioned by season and league: `fbref` (10,355 player-seasons), `fbref_raw` (18,243), `understat_team` (684 club-seasons), `tm_transfers` (168,248), `tm_valuations` (615,190), `tm_players` (50,149), plus two derived tables: league minutes per player, club and season, and league tables rebuilt from match results. The rebuilt tables agree with the mid-sem table on 99.1% of club-seasons. Writing the lake takes 22 s in local mode. A query for one season and league reads one partition (the plan in notebook 01 shows the partition filter).

Entity resolution. FBref has no player id. Pass one learns the club alias map from the data: join on season, league, birth year and exact name key, then give each FBref club the Transfermarkt club most of its players belong to. All 145 clubs map one to one. Pass two blocks on (season, club) and compares accent-insensitive, token-sorted name keys with Spark `levenshtein` and a Jaro-Winkler pandas UDF, adds birth year and position group, and keeps mutual best pairs. It links 16,474 of 16,865 outfield player-club-seasons (97.7%), and 99.1% of those with 900 minutes or more; the worst league-season is 98.1%. 33 of 2,751 feeder-seasons stay unmatched and are listed in `outputs/unmatched_feeders.csv`.

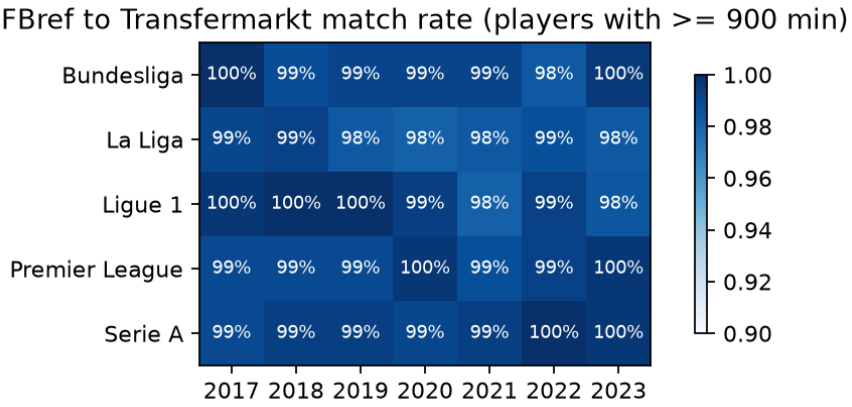


Figure 2: Match rate per league and season for players with at least 900 minutes.

Replacement events. An event is an elite club whose starter (1,350 minutes or more) plays under 450 league minutes for it the next season, and which signs a player of the same fine position in the summer between, who then plays at least 450 minutes. Loans are removed by walking each player's moves in date order (23,044 loan departures found). This gives 464 events from 371 signings. Only 23.2% of those signings (86) were inside the candidate pool (age 26 or under, 900 minutes, non-elite club in the five leagues). 24.3% came from clubs outside the five leagues, such as Benfica or Feyenoord, and the rest were older, at another elite club, or had played too little. The engine can only be right about the 95 events whose signing was in the pool, and all results below are restricted to them.

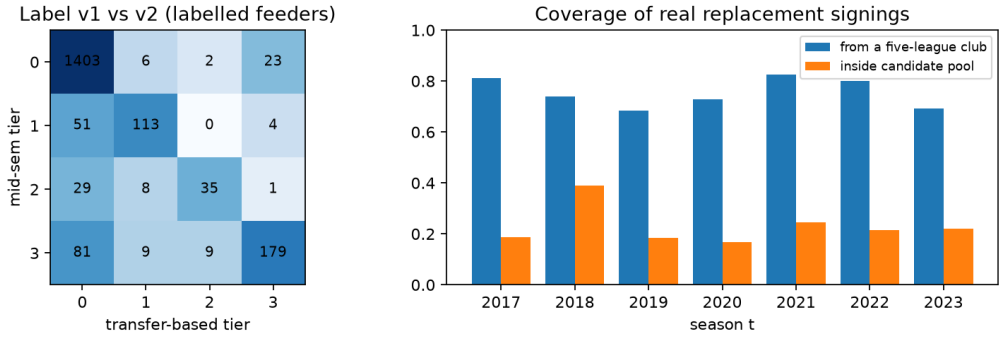


Figure 3: Left: mid-sem label against the transfer label. Right: share of real replacement signings that came from a five-league club and that were inside the candidate pool.

3 Post-mid-semester topics: one new algorithm per task, in PySpark

None of these was used before the mid-sem. Each runs in Spark (local mode) with its plan shown once in notebook 03.

Task	Algorithm (Spark)	Result	Time (s)
Similarity search	Euclidean LSH, MinHash LSH	95% of true top-25 neighbours kept while comparing 88% of pairs; both views together keep 95% of real signings	0.4
Sequential patterns	PrefixSpan	477 frequent career sequences, 176 with lift $\geq$ 1.3; flagged players move up 27.5% vs 25.0% out of time	1.0
Clustering	Bisecting k-means per fine position	23 archetypes, mean silhouette 0.17; Cramer's V with moving up 0.20	56
Link analysis	GraphFrames PageRank; power-iteration clustering	pipeline rank alone AUC 0.573 (sales count 0.579); communities with modularity 0.32	22
Recommendation	ALS, implicit feedback	MRR 0.631 vs popularity 0.697: ALS loses	13
Classification	LinearSVC, multilayer perceptron	LinearSVC AUC 0.687, MLP 0.558 (mid-sem best 0.681)	147
Outlier detection	k-NN distance score (self-join)	hidden-gem flag: 15.0% move up vs 22.2%	4.4
Stream mining	Structured Streaming, Count-Min Sketch	live team dominance reaches r = 0.9 with the season value after about 12 matches	10

Table 1: The eight Spark mining tasks.

Similarity search (LSH). Random-projection LSH hashes vectors so that close ones tend to share a bucket, and only bucket-mates are compared. It generates the candidates for each event from style-neutral vectors; MinHash on top-quartile trait sets is the second view. The honest reading is that the pool is small (about 96 same-position players per event), so LSH saves little here: the chosen setting (4 hash tables, bucket length 2) still compares 88% of pairs. It is in the pipeline because the same code scales to a pool of every league in the world.

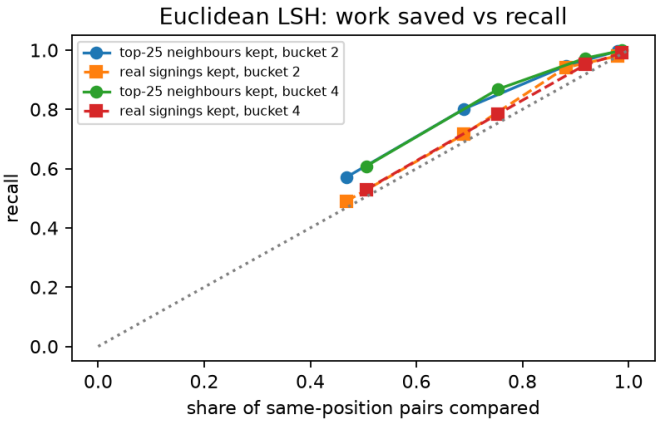


Figure 4: LSH: share of pairs compared against recall of the exact top-25 neighbours and of the real signings.

Sequential patterns (PrefixSpan). A career is an ordered list of seasons, each an itemset of club tier, league, role and form band. PrefixSpan mines the frequent orders on feeder-seasons up to 2019. The strongest multi-season pattern is {lg=SA} → {club=mid, form=mid} (lift 2.34 in training, 1.06 later). Out of time the flagged group moves up at 1.10 times the base rate (Fisher p = 1.7e-07): a real but small signal, and the train and test lifts of individual patterns correlate only 0.26.

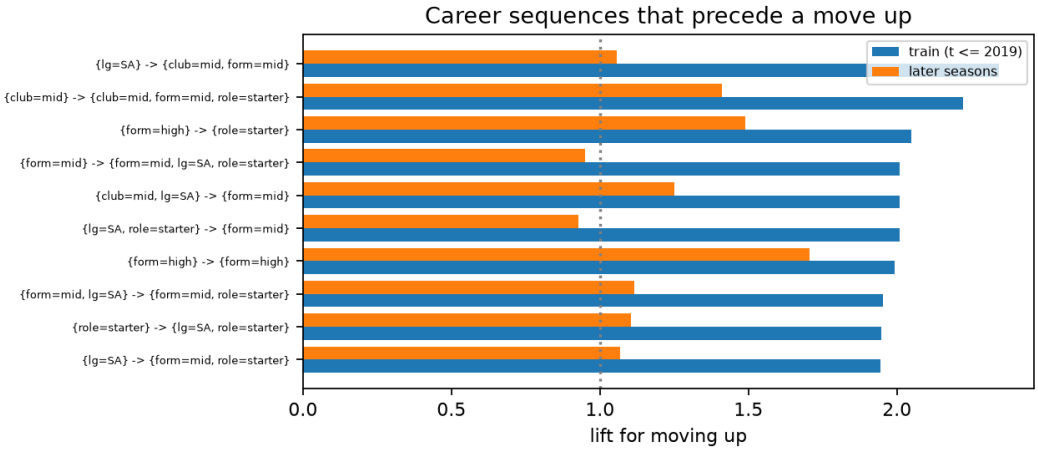


Figure 5: Top career sequences by lift for moving up, in training and in later seasons.

Clustering (bisecting k-means). Bisecting k-means splits the worst cluster in two repeatedly. It is run inside each of the seven Transfermarkt fine positions, which the mid-sem could not do with three coarse groups. It finds 23 archetypes; archetype and moving up are associated (chi-square p = 2.9e-14, Cramer's V 0.20). Creative archetypes are bought more often than defensive ones at every position.

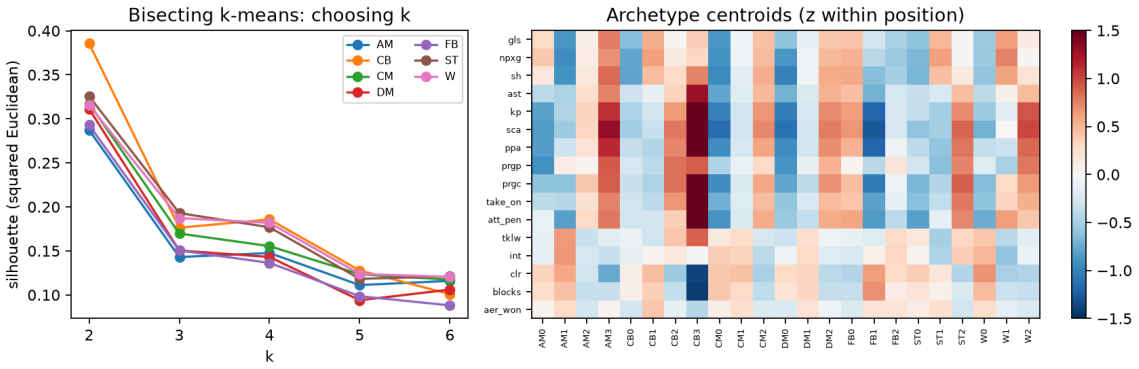


Figure 6: Silhouette by k for each position, and the archetype centroids.

Link analysis (PageRank, PIC). Each permanent transfer is an edge (61,230 in all). With edges pointing from buyer to seller, PageRank is high for clubs whose players are bought by important clubs. Academies and one-off amateur clubs had to be removed first, because the reversed graph rewards them. The top of the 2023 list is Salzburg, Indep. Medellín, Olympiacos, Benfica, Boca Juniors. For feeders, the club's pipeline rank alone gives AUC 0.573, which is no better than simply counting the club's past sales (0.579). Power-iteration clustering on 1,001 clubs gives transfer communities with modularity 0.32; the iteration count has to be chosen (here by modularity), because too many iterations collapse every club into one group.

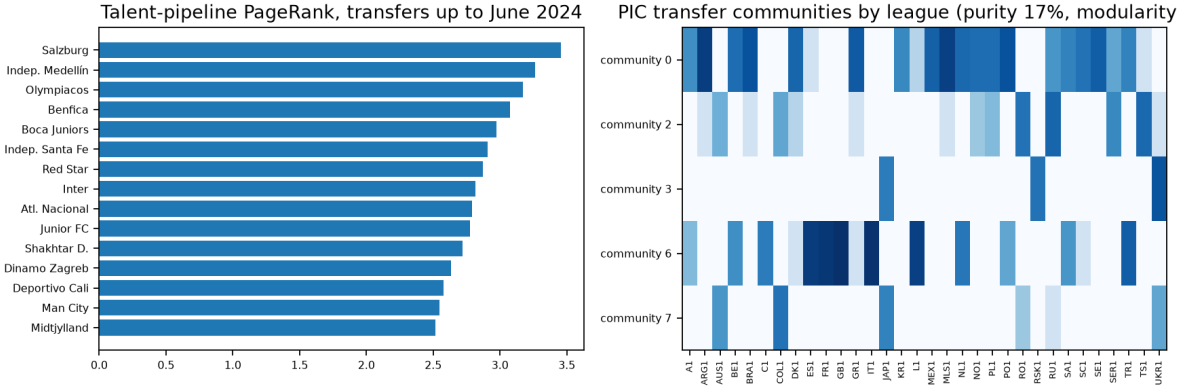


Figure 7: Left: clubs with the highest talent-pipeline PageRank. Right: transfer communities by league.

Recommendation (ALS). ALS factorises the club by archetype matrix of past signings. This did not work. On 1,119 later signings, ALS ranks the bought archetype worse than league-wide popularity (MRR 0.631 against 0.697; difference 95% interval [-0.085, -0.047]). The training matrix has only 364 signings across 132 clubs, too few to learn a club's taste. The score is kept in the engine so the ablation can measure it.

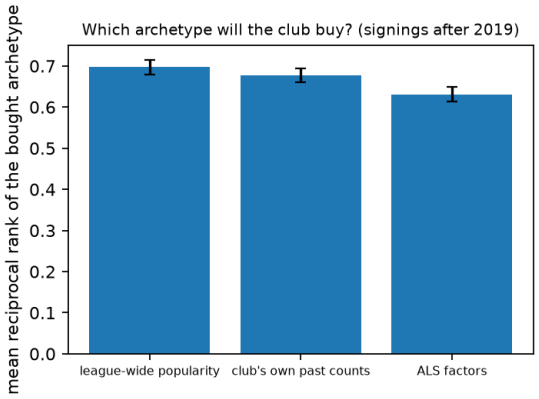


Figure 8: Mean reciprocal rank of the archetype a club actually bought.

Classification (LinearSVC, MLP). Same question and same walk-forward split as the mid-sem classifiers. LinearSVC reaches AUC 0.687 with 41 hits in 75 picks, level with the mid-sem logistic regression (0.681, 46 hits). The perceptron overfits the 700 to 1,900 training rows and is clearly worse (AUC 0.558). On the transfer label over five test seasons LinearSVC gets AUC 0.680 and 62 hits in 125.

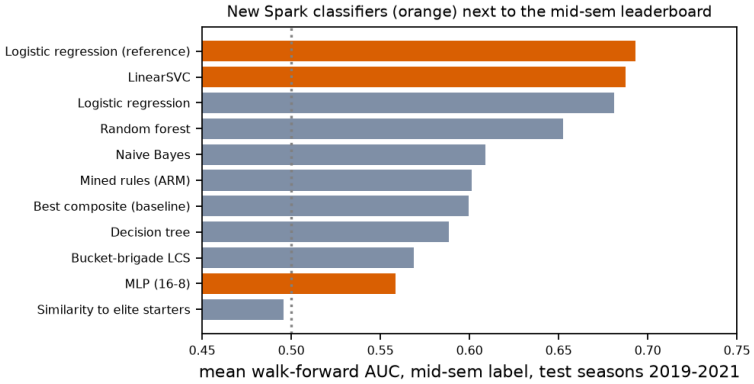


Figure 9: New Spark classifiers next to the mid-sem leaderboard (same label, same test seasons).

Outlier detection (k-NN distance). The score is the mean distance to the ten nearest neighbours inside the archetype. Unusual players are bought slightly more often (Spearman -0.088, p = 0.000). The hidden-gem flag (top decile of the score and above-median composite index) marks 100 feeders who moved up at 15.0% against 22.2% for the rest (Fisher p = 9.7e-01).

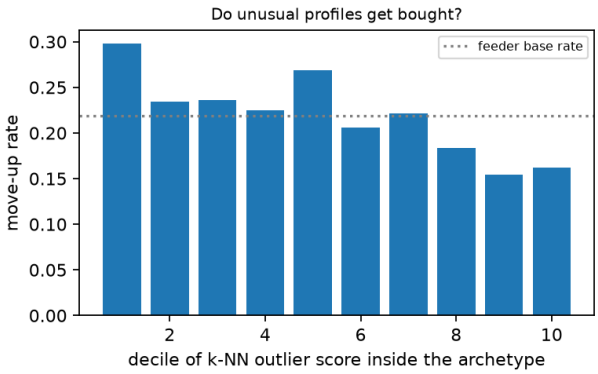


Figure 10: Move-up rate by decile of the k-NN outlier score.

Stream mining. The 2023-24 Understat matches are replayed one week per micro-batch (39 batches). A stateful aggregation keeps each team's running dominance; after five weeks it correlates 0.80 with the end-of-season value and passes 0.9 at about 12 matches per team. A Count-Min Sketch counts deep completions per team in fixed memory: at width 200 the mean relative error is 0.0%, and at width 20 it over-counts by 2.5 times the true count, which shows what the width buys.

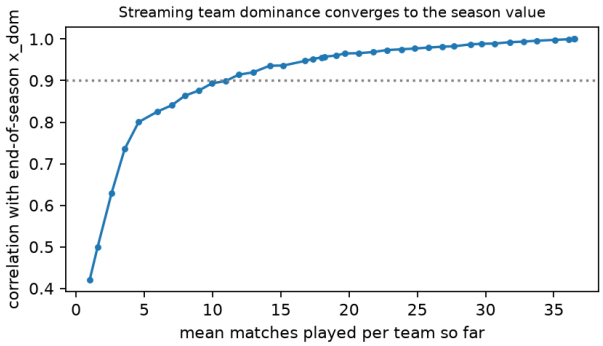


Figure 11: Streaming team dominance against the end-of-season value.

4 Techniques beyond the course

Survival analysis. Feeder-seasons become person-period rows (2,562 rows, 314 events), which lets the last two seasons be used (697 rows). The discrete-time hazard is a Spark logistic regression refitted each season on past rows only; its walk-forward AUC is 0.688 and its predicted and observed rates differ by 2.2% on average across deciles. On 819 later entrants its concordance index is 0.675; a Cox proportional-hazards model (`lifelines`) fitted as a check gets 0.656.

Learning to rank. LightGBM `LGBMRanker` (LambdaMART) with one query per replacement event, against a pointwise Spark `GBTClassifier`. Section 6 has the results.

Conformal prediction. Split-conformal turns a score into a list with a stated coverage. For movers: to catch 90% of next season's real movers the hazard list must hold 84% of all feeders, and it then caught 94.5% on later seasons; a 50% promise needs 26% of feeders and caught 43.8%. For replacement events, calibrated on 26 events and checked on 38: a list that contains the real signing 90% of the time needs 70% of the pool (about 70 players) and covered 86.8%; a 50% promise needs about 32 players and covered 57.9%. The guarantees hold approximately, and they show that a short list cannot promise much.

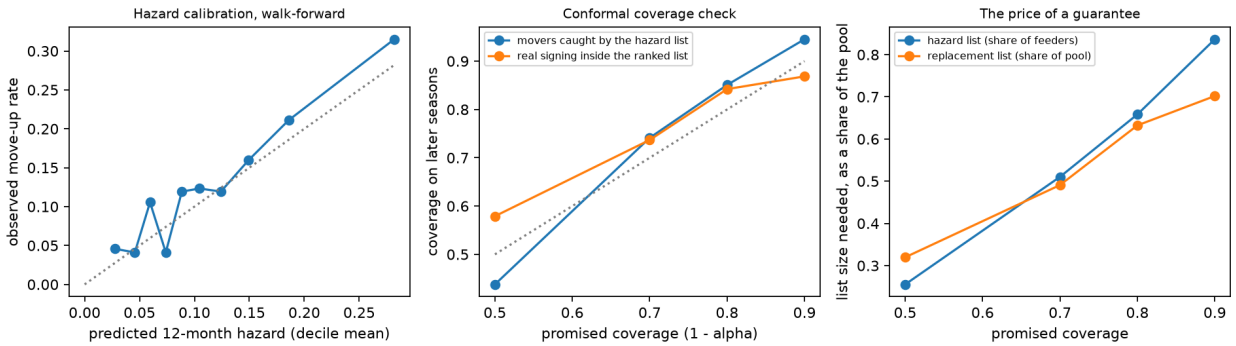


Figure 12: Hazard calibration, conformal coverage on later seasons, and the list size a guarantee costs.

SHAP (TreeExplainer on the ranker) explains each candidate's rank, and **dynamic time warping** is compared with the simple trajectory similarity: the two correlate 0.43 and put the real signing

in the top 10 equally often (17% and 17%), so DTW adds nothing here. Only 14% of pairs could be age-aligned, because the starter's seasons at the candidate's age are usually before 2017.

5 Style translation and its check on real movers

The mid-sem removed team style from a player's numbers. Translation runs it the other way, into the buying club:

$$\text{stat in } C = \text{stat}_{\text{neutral}} \times \exp\Big(\sum_{\text{supported terms}} \beta_{\text{pos,metric,term}}\, x_{\text{term}}(C,t) \Big) \times \text{league factor}(C)$$

using the mid-sem elasticities and league offsets unchanged. The check uses 1,110 players who changed club between consecutive seasons (460 across leagues). Their per-90 numbers at the new club are predicted from (a) last season's raw numbers, (b) style-neutral numbers, and (c) neutral numbers translated into the new club, using the club's style from the season before the move.

Mean absolute error, in standard deviations of each metric: (a) 0.495, (b) 0.486, (c) 0.489. Translation beats raw numbers by a small amount (difference [-0.010, -0.003], 95% bootstrap) and in 37 of 48 position-by-metric cells. It does not beat the plain style-neutral numbers (difference [-0.001, +0.006]). Most of the gain is from removing the old club's style; adding the new club's style contributes nothing measurable, even with the new club's true style that season (0.484). So translation is kept as a display feature (the radar) and as one ranking score, with the gain over raw numbers stated at about 1.2% of the error.

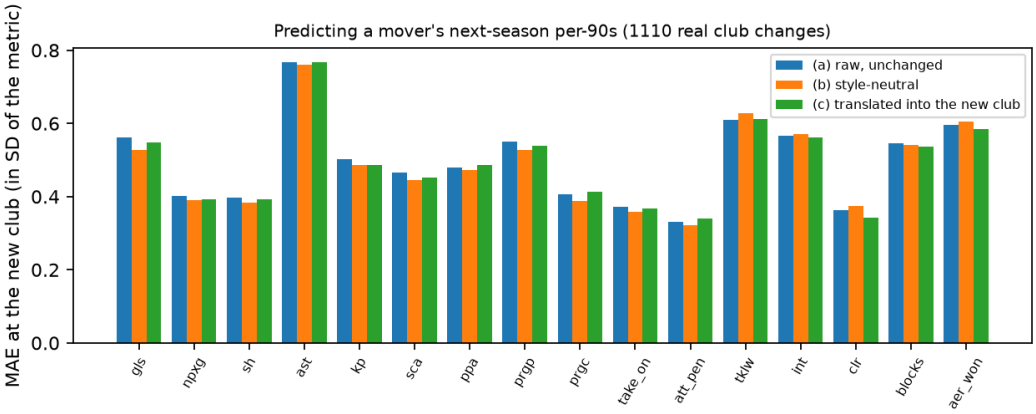


Figure 13: Error in predicting a mover's next-season per-90s, by metric.

6 Results

For each event the engine scores every same-position pool player on translated similarity, trajectory similarity, the selling club's pipeline PageRank, the buying club's ALS taste, the survival hazard, and market value (level, and gap to the departing starter), with age and composite index. A LambdaMART ranker combines them. It is trained walk-forward: events of season Y are ranked by a model that saw only events up to $Y-1$. The test set is 64 events from 2020-21 to 2023-24, with about 101 candidates each.

Method	Median rank	R@5	R@10 [95% CI]	R@25	R@50	p vs best young
Random order of the pool	46.0	6%	12% [5, 20]	28%	55%	
Highest market value	19.2	14%	27% [16, 38]	62%	78%	0.27
Raw cosine similarity to the starter	35.0	12%	25% [14, 36]	36%	62%	0.36
Best young performer (composite index)	29.5	5%	17% [8, 27]	41%	66%	
Single score: hazard	22.5	17%	28% [17, 39]	56%	78%	0.06
Single score: translated similarity	36.5	14%	22% [12, 33]	31%	62%	0.63
Shadow Squad fusion with Spark GBT (pointwise)	34.5	2%	14% [6, 23]	38%	70%	0.81
Shadow Squad (LambdaMART, all scores)	26.0	12%	31% [20, 42]	47%	73%	0.08
Ablation: no translation	28.5	6%	19% [9, 28]	47%	73%	1.00
Ablation: no trajectory	25.5	11%	25% [14, 36]	50%	75%	0.37
Ablation: no PageRank	26.5	17%	25% [16, 36]	48%	73%	0.36
Ablation: no ALS taste	26.0	14%	23% [14, 34]	48%	75%	0.48

Table 2: Replacement recall@K and median rank of the real signing on 64 unseen in-pool events. Intervals are bootstrap over events; the last column is a paired permutation test on recall@10 against the best-young-performer baseline.

What worked. Shadow Squad puts the real signing in the top 10 for 31% of events. A random order gets 12%, the mid-sem method (raw cosine similarity) 25%, and the mid-sem result to beat (best young performer) 17%. That is the highest recall@10 in the table.

What did not. With 64 events the difference from the best-young-performer baseline is not statistically significant ($p = 0.08$; difference in recall@10 [+0.00, +0.28]). Two single scores do about as well as the full engine: the highest market value has a better median rank (19.2 against 26.0) and better recall@25 (62% against 47%), and the hazard alone reaches 28% at 10 and 56% at 25. Elite clubs mostly sign the players everyone already values. The pointwise Spark GBT is worse than LambdaMART (14% at 10).

Ablation. Removing one score at a time changes recall@10 from 31% to 19% without translation, 25% without trajectory, 25% without PageRank and 23% without ALS taste. Every interval overlaps the full model's, so no single score can be said to matter on this many events. Translation shows the largest drop.

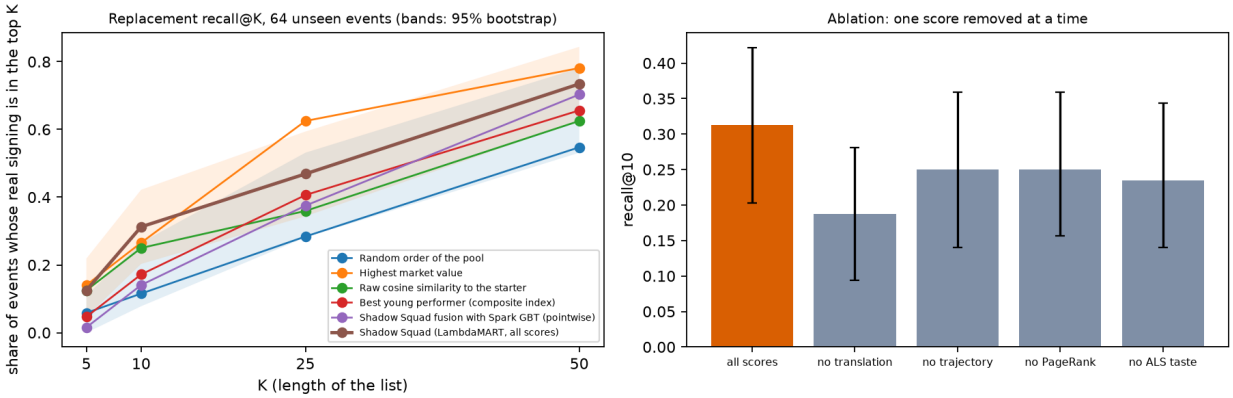


Figure 14: Left: recall@K with 95% bootstrap bands. Right: recall@10 with one score removed at a time.

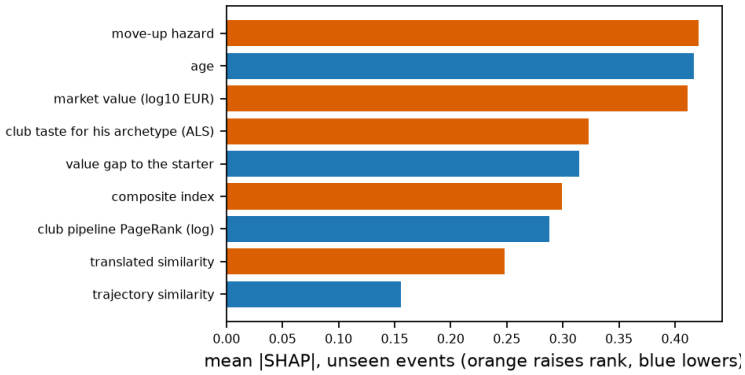


Figure 15: Mean absolute SHAP value of each score on unseen events.

7 Model predictions, with an explanation for each

The screenshots are from the Streamlit app (`app/shadow_squad.py`), which reads the Parquet lake: choose a league, an elite club, a season and a departing starter. It shows the ranked list, what the club really did, the radar (starter's actual numbers, candidate raw, candidate translated into the club), the SHAP waterfall for the chosen candidate, and market values. The cases were picked by rule: the three unseen events with the best rank for the real signing, and the unseen event where a real signing was ranked worst.

8 Case studies

Three hits

Dortmund 2023-24: Niclas Füllkrug (ST) replaced by Maximilian Beier from Hoffenheim (fee EUR 28.5m). The engine ranked him 2 of 76 pool players at that position. His strongest scores were move-up hazard (higher than 91% of the pool), market value (higher than 88% of the pool), club pipeline rank (higher than 86% of the pool). The first name on the list was Elye Wahi (Lens).

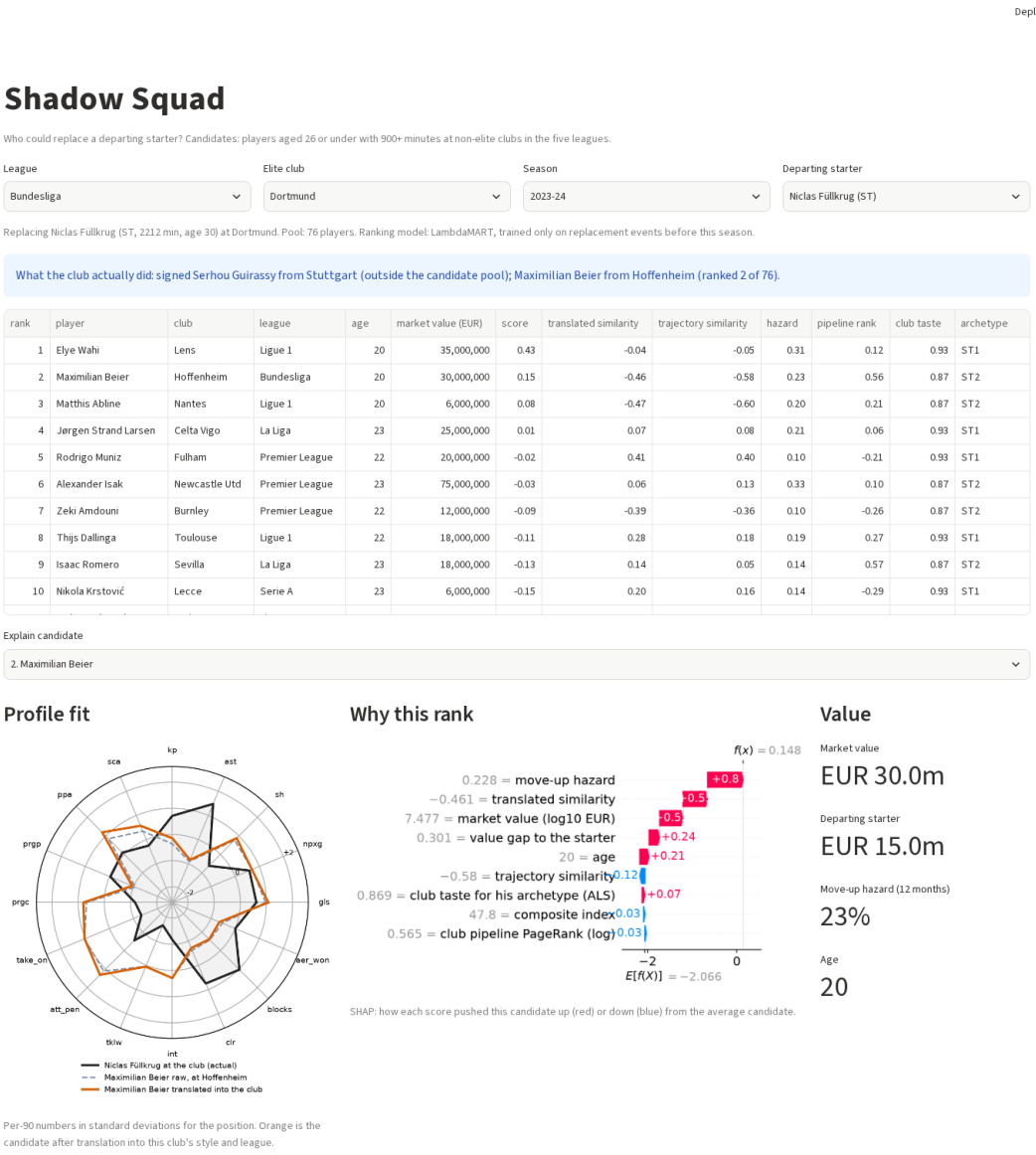


Figure 16: Dashboard for this event: ranked list, translated radar and SHAP waterfall for Maximilian Beier.

Paris S-G 2022-23: Neymar (AM) replaced by Kang-in Lee from RCD Mallorca (fee EUR 22.0m). The engine ranked him 2 of 42 pool players at that position. His strongest scores were move-up hazard (higher than 95% of the pool), translated similarity (higher than 93% of the pool), trajectory similarity (higher than 81% of the pool). The first name on the list was Gabriel Veiga (Celta Vigo).

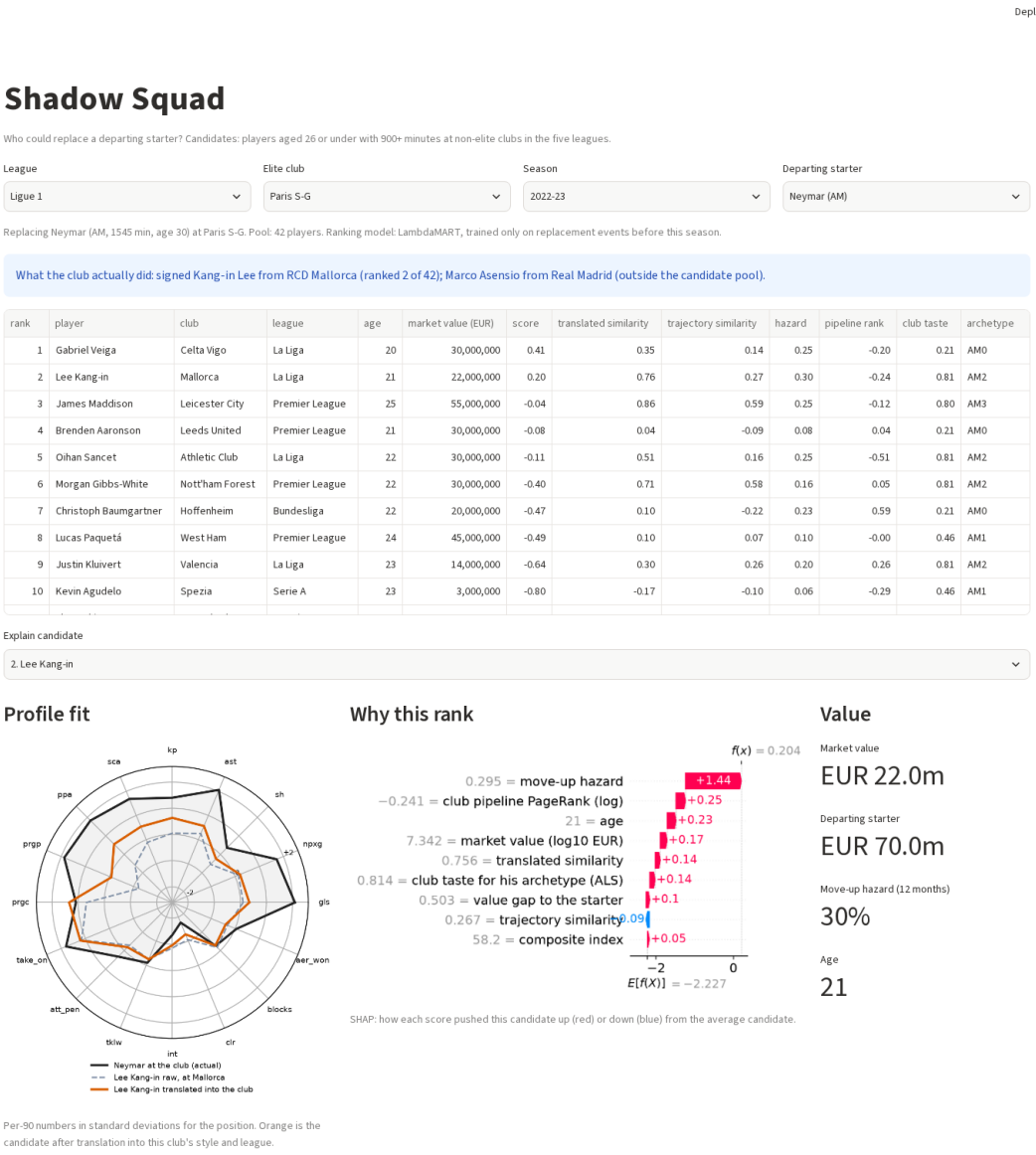


Figure 17: Dashboard for this event: ranked list, translated radar and SHAP waterfall for Kang-in Lee.

Paris S-G 2021-22: Georginio Wijnaldum (CM) replaced by Carlos Soler from Valencia (fee EUR 18.0m). The engine ranked him 2 of 104 pool players at that position. His strongest scores were move-up hazard (higher than 98% of the pool), market value (higher than 97% of the pool), composite index (higher than 75% of the pool). The first name on the list was Pablo Fornals (West Ham).

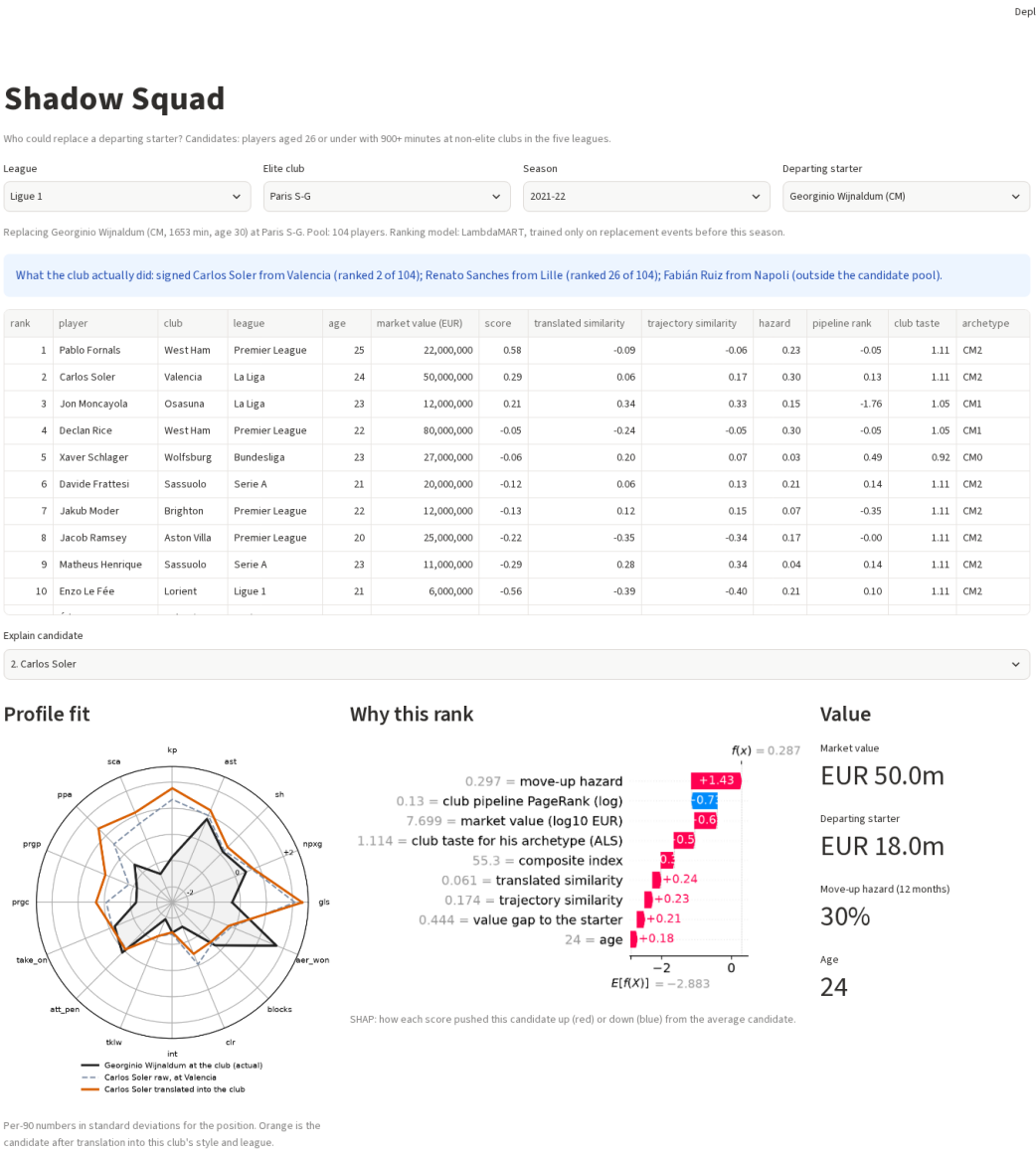


Figure 18: Dashboard for this event: ranked list, translated radar and SHAP waterfall for Carlos Soler.

One miss

Marseille 2022-23: Sead Kolašinac (CB) replaced by Bamo Meïté from FC Lorient (fee EUR 10.5m). The engine ranked him 131 of 133, close to last. His weakest scores were market value (higher than 2% of the pool), age (higher than 2% of the pool), composite index (higher than 8% of the pool). His market value was EUR 1.5m against EUR 5.0m for the starter he replaced. The ranker has learned that elite clubs mostly buy established, highly valued players, so a cheap squad signing with a thin record is exactly the kind of move it cannot see coming.

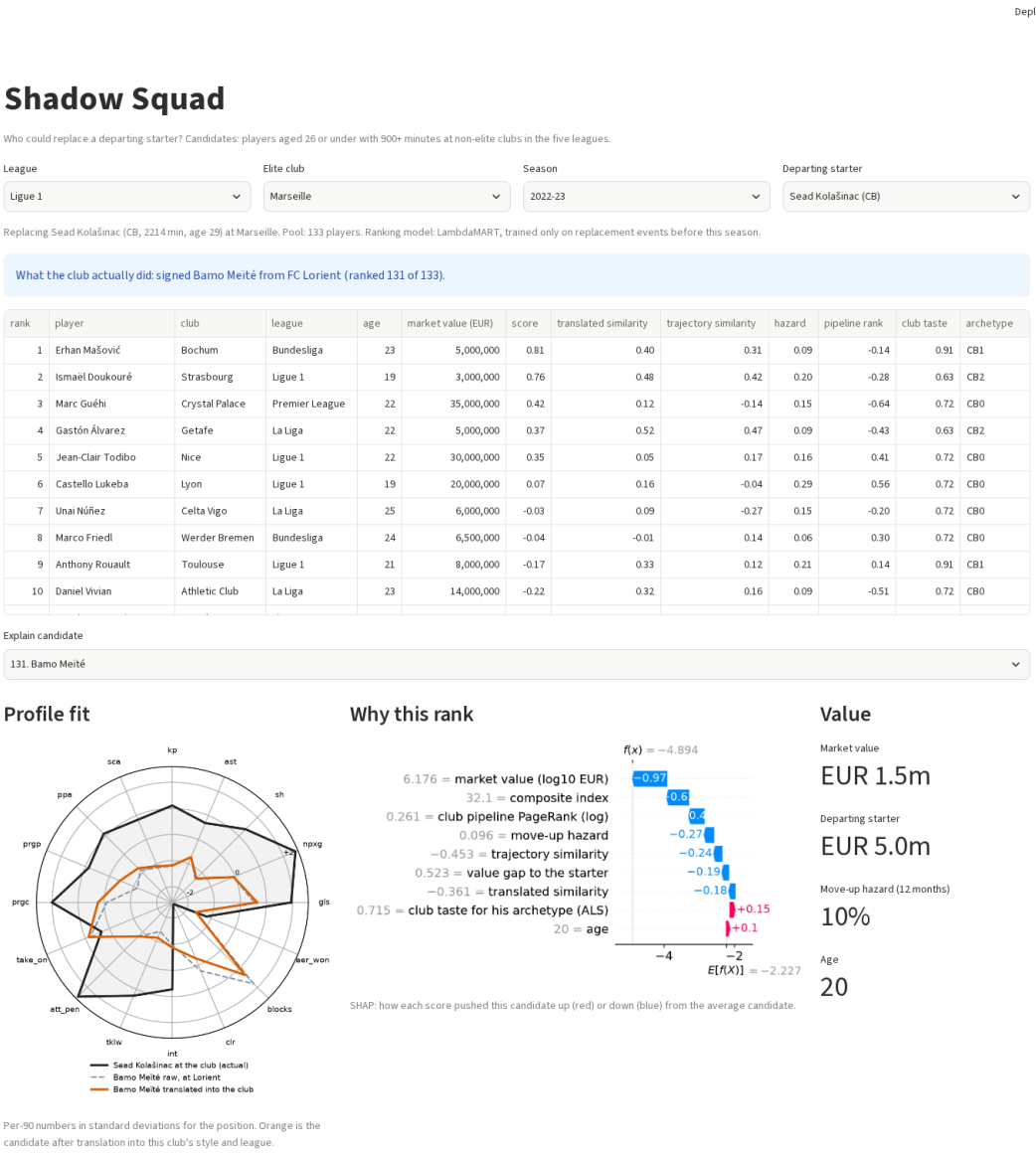


Figure 19: Dashboard for this event: ranked list, translated radar and SHAP waterfall for Bamo Meïté.

9 Limitations and conclusion

- Coverage. 77% of real replacement signings were outside the candidate pool, most of them from leagues the data does not cover. The engine is evaluated only on the rest.
- Sample size. 95 in-pool events in seven seasons, 64 of them unseen. The intervals are wide and the method order in the results table is not reliable.
- Transfer records. 9.1% of feeder-seasons belong to players with no record in the Transfermarkt dump and are labelled ”no move”. Loans are inferred from move patterns, not from a loan flag.
- The style elasticities, league offsets, archetypes and outlier scores are fitted on all seasons without outcome labels, as in the mid-sem. The ranker, hazard, PageRank and ALS scores

use only past data.

- Local Spark. Everything runs in local mode on one machine; the data is far smaller than what Spark is for.

Reframing the task as "who replaces whom" gives a test that the mid-sem lacked: a real decision by a real club. On that test the engine has the best top-10 recall of the methods tried and nearly doubles the mid-sem baseline, but the margin is not significant on 64 events, and market value alone is as good or better deeper in the list. Style translation improves on raw numbers by about 1.2% and no more than style-neutral numbers do. The parts that held up are the transfer-based labels, the entity resolution, the survival hazard, and a dashboard that shows why each candidate is ranked where he is.